Syllabus Details

Syllabus ID:	12223
Syllabus Name:	**Programming Fundamentals_Cơ sở lập trình**
Course Name English:	**Programming Fundamentals**
Subject Code:	**PRF192**
Learning-Teaching Method:	Problem based learning, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	None
Description:	Understand basics computer system and methods of software development, focus on function-oriented programming design, coding, testing and discipline in programming. Explain basic concepts of programming, function-oriented programming design, modularity, understand and coding programs using C language. Upon completing the course, student should have: 1. Knowledge: (ABET e) - Explain the way to solve a real problem using computer. - Understand the basic concepts computer system, and software development. - Understand the basic concepts of programming, focus on procedure programming, testing and debugging, unit testing. 2. Skills in programming: (ABET k) - Read and understand the simple C programs; - Solve real problems using C. 3. Apply learning methods effectively: (ABET i) - Academic reading. - Individual and team work behaviors.
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all assigned exercises given by instructor in class or at home and submit on time. - Use laptop in class only for learning purpose. - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information.
Tools:	Tools: - Internet - C language utility (ex.DevC++ 6.3)
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1286/QĐ-ĐHFPT dated 11/22/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/22/2024

3 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Foundations of Programming Using C	Evan Weaver	Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported License	2003			True	True	False	
2	The C Programming Language	Brian W. Kernighan, Dennis M. Ritchie	Pearson	1988	2nd Edition	978-0131103627	False	True	False	
3	MOOC: Introduction to C Course	Chris Szalwinsk	Chris Szalwinski and Seneca College.		N/A	N/A	False	False	True	https://intro2c.sdds.ca/

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand and describe about a program, and the way it works on computer.
2	CLO2	Demonstrate about variables, expressions and basic operations and illustrate them by program examples.
3	CLO3	Explain the meaning and use of Logic constructs in C Language. Describe about programming styles.
4	CLO4	Describe about Modularity and Functions in C programs and their use.
5	CLO5	Understanding and using pointers in programs.
6	CLO6	Discuss about C Libraries and their use.
7	CLO7	Discuss about Arrays, Struct and their use in C programs.
8	CLO8	Discuss about Strings and their use in C programs.
9	CLO9	Discuss about Files and their use in C programs.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Introduction and installing tools	Offline	CLO1	I	Textbook	PRF192_Slides_1	Read Textbook, install tools	
2	Basic concepts computer system; steps to develop a software	Offline	CLO1	T	Textbook	PRF192_Slides_2	Read Textbook	
3	Structure of a simple C program; C compiler	Offline	CLO1	T	Textbook	PRF192_Slides_3	Practice a simple C program	
4	Variable and constant used in the C programming language	Offline	CLO2	T	Textbook		Practice declaration variable and constant	
5	Understand what is a data type	Offline	CLO2	T	Textbook		Read Textbook	
6	Basic Memory Operations Implement basic input/output functions using scanf, printf	Offline	CLO2	T	Textbook		Read Textbook Practice basic input/ ouput	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
7	Expressions: Arithmetic, Relational, Logical, Bit operators, Shorthand Assignment operators	Offline	CLO2	T	Textbook		Implement operators	
8	Expressions: Mixing Data Types, Casting, Precedence	Offline	CLO2	T	Textbook		Practice Mixing data types and Casting	
9	Structured Programming: Selection (if-else, switch-case)	Offline	CLO2	T	Textbook		Implement if-else, switch-case	
10	Structured Programming: Loop (for, while, do-while)	Offline	CLO3	I	Textbook		Implement for, while, do-while	
11	Review: The lecturer reviews the learned content	Offline	CLO1, CLO2, CLO3	T	Textbook		Answer the Instructor's questions and practice the requirements	
12	Review (contd): The lecturer reviews the learned content	Offline	CLO1, CLO2, CLO3	T	Textbook		Answer the Instructor's questions and practice the requirements	
13	Workshop 1: Practice declaring and initializing values for variables.	Offline	CLO1, CLO2, CLO3	U	Completed Workshop		Complete Workshop 1	
14	Workshop 1: Use expressions to perform simple calculations.	Offline	CLO1, CLO2, CLO3	U	Completed Workshop		Complete Workshop 1	
15	Workshop 1: Practice techniques to control program structure using conditional and loop structures	Offline	CLO1, CLO2, CLO3	U	Completed Workshop		Complete Workshop 1	
16	Modules: What is a module? How to design modules for a program?	Offline	CLO4	I	Textbook		Read Textbook	
17	Modules: Characteristics of modules	Offline	CLO4	I	Textbook		Read Textbook	
18	Modules: Principles of defining modules	Offline	CLO4	I	Textbook		Read Textbook	
19	Functions: Function definitions	Offline	CLO4	T	Textbook		Practice function definitions	
20	Functions: Use a function. Parameters: Pass by value	Offline	CLO4	T	Textbook		Practice use a function	
21	Functions: Differentiate Built-in and User-defined functions	Offline	CLO4	T	Textbook		Read Textbook	
22	Functions: Extent and Scope of a Variable	Offline	CLO4	T	Textbook		Implement scope of variable	
23	Review: The lecturer reviews the learned content	Offline	CLO4	T	Textbook		Answer the Instructor's questions and practice the requirements	
24	Review (cont): The lecturer reviews the learned content	Offline	CLO4	T	Textbook		Answer the Instructor's questions and practice the requirements	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
25	Workshop 2: Real-world problem analysis techniques for program modularization	Offline	CLO4	U	Textbook		Complete Workshop 2	
26	Workshop 2:Practice defining and using functions	Offline	CLO4	U	Textbook		Complete Workshop 2	
27	Workshop 2: Build a program in menu form to call functions	Offline	CLO4	U	Textbook		Complete Workshop 2	
28	Introduction to Pointers. Where are pointers used?	Offline	CLO5	I	Textbook		Read Textbook	
29	Pointer Declarations; Pointer operators	Offline	CLO5	T	Textbook		Practice pointer declarations and Implement pointer operators	
30	Pointers as parameters of a function. Pass by reference	Offline	CLO5	T	Textbook		Practice pass by reference to parameters of a function	
31	Pointers: Dynamic Allocated Data	Offline	CLO5	T	Textbook		Practice dynamic allocated data	
32	Libraries: Standard - stdlib.h; Time - time.h	Offline	CLO6	I	Textbook		Read Textbook	
33	Libraries (cond): Math - math.h; Character - ctype.h	Offline	CLO6	T	Textbook		Read Textbook	
34	Input/ Formatted Output	Offline	CLO6	T	Textbook		Implement input and formatted output	
35	Review: The lecturer reviews the learned content	Offline	CLO5, CLO6	T	Textbook		Answer the Instructor's questions and practice the requirements	
36	Progress Test 1: Evaluate students on the theoretical knowledge they have learned	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U	Study Materials		Complete Progress Test 1	
37	Arrays: Conceps One-Dimensional array and Basic Operations on Array	Offline	CLO7	T	Textbook		Practice one-demensional array declarations and basic operations on array: input/ output	
38	Matrix: Two-Dimensional Arrays; Inputting, Outputting Data for Matrix and other operations	Offline	CLO7	T	Textbook		Practice two-demensional array declarations, input/output, operations	
39	Arrays: Solve Problems using Arrays (Linear Search and Selection Sort)	Offline	CLO7	T	Textbook		Practice: Search, Sort	
40	Struct: Using struct to organize complex data	Offline	CLO7	T	Textbook		Practice declaration, use Struct	
41	Review: The lecturer reviews the learned content	Offline	CLO7	T	Textbook		Answer the Instructor's questions and practice the requirements	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
42	Review: Array or structs	Offline	CLO7	T	Textbook		Practice case study	
43	Workshop 3: Practice basic input/ output and operations of array	Offline	CLO7	U	Completed Workshop		Complete Workshop 3	
44	Workshop 3: Practice with character arrays using library functions in ctype.h	Offline	CLO6, CLO7	U	Completed Workshop		Complete Workshop 3	
45	Workshop 3: Manage data with an array of struct	Offline	CLO7	U	Completed Workshop		Complete Workshop 3	
46	String: Declare/Initialize a string	Offline	CLO8	T	Textbook		Practice declare/ initialize String	
47	String: Input/ Output a String using buit-in Functions	Offline	CLO8	T	Textbook		Practice input/ output and operations string	
48	Array of strings; operations of String (string.h)	Offline	CLO8	T	Textbook		Implement Array of string	
49	Review: The lecturer reviews the learned content	Offline	CLO8	T	Textbook		Answer the Instructor's questions and practice the requirements	
50	Review (contd): The lecturer reviews the learned content	Offline	CLO8	T	Textbook		Answer the Instructor's questions and practice the requirements	
51	Workshop 4: Problem-Solving with String (input/ output/ operations)	Offline	CLO6, CLO7, CLO8	U	Completed Workshop		Complete Workshop 4	
52	File: Basic concepts File	Offline	CLO9	I	Textbook		Read Textbook	
53	File: File types (Text File, Binary File)	Offline	CLO9	T	Textbook		Read Textbook	
54	File: Ways for accessing files	Offline	CLO9	T	Textbook		Implement acessing files	
55	File: File Functions	Offline	CLO9	T	Textbook		Practice file functions	
56	Review: The lecturer reviews the learned content	Offline	CLO9	T	Textbook		Answer the Instructor's questions and practice the requirements	
57	Workshop 5: Open, read, write, and close files using C file handling functions	Offline	CLO9	U	Completed Workshop		Complete Workshop 5	
58	Progress Test 2	Offline	CLO7, CLO8, CLO9	U	Study Materials		Complete Progress Test 2	
59	Assigment	Offline	CLO1 - CLO9	U	Completed Assignment		Complete Assignment	
60	Final Review	Offline	CLO1 - CLO9	U	Study Materials		Continue exam preparation	

6 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	What is C programming language, and what are its primary uses?
2	1	CQ1.2	What are the basic structure and syntax of a C program?

No.	Session No	Name	Details
3	1	CQ1.3	What does the #include preprocessor directive do?
4	2	CQ2.1	What is the purpose of the main() function in a C program?
5	2	CQ2.2	How do you comment your code in C?
6	2	CQ2.3	What is the significance of semicolons in C?

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	15.0%	>0	at home	CLO2-CLO9	Option 1: a problem similar to real one; Option 2: Follow lecturer's proposal	Option 1: N/A; Option 2: Follow lecturer's proposal	problem(s) to solve by programming using CLO2 - CLO9	Individual or team work, guided by instructor, submission by a given deadline	
2	Practical Exam	on-going	1	30.0%	>0	85'	CLO2-CLO8	Preferable to be marked by scripts	1-2 problems	problem(s) to solve by programming using CLO2 - CLO8	supervised by proctor(s) sent by exam board	Automatic or manual evaluating
3	Progress test	on-going	2	15.0%	>0	20'-40'	CLO1, CLO5, CLO6, CLO9	Option 1: essay or multiple choice; Option 2: Follow lecturer's proposa	Option 1: 15-30 if multiple choice; Option 2: Follow lecturer's proposal	PT1: CLO1 - CLO5; PT2: CLO6 - CLO9; Number of questions: 3-8 for each CLOx	supervised by instructor	Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded.
4	Workshop	on-going	5	10.0%	>0	on class room	CLO1-CLO9	Option 1: practical exercises; Option 2: Follow lecturer's proposal	Option 1: N/A; Option 2: Follow lecturer's proposal	W1: CLO1, CLO2, CLO3; W2: LO4; W3: CLO6, CLO7; W4: CLO7, CLO8; W5: CLO9	Teachers support tutorials and students do them by themselves	A workshop will be evaluated at the beginning of the next lesson

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
5	Final exam	Final exam	1	30.0%	4	60'	CLO1-CLO9	multiple choice	50	questions distribution:CLO1: 2 - 3; CLO2: 5 - 10; CLO3: 8 - 10; CLO4: 8 - 10; CLO5: 5 - 8; CLO6: 5 - 8; CLO7: 5 - 8; CLO8: 2 - 4. More than 70% new questions (for the current semester);	supervised by proctor(s) sent by exam board	